

RETAIL SALES ANALYTICS PROJECT REPORT

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Executive Summary

This project leveraged SQL Server and Power BI to transform raw retail sales data into actionable business intelligence. The workflow included data cleaning, exploratory data analysis, KPI development, and dashboard design to evaluate revenue performance, profitability, customer behavior, regional performance, and operational efficiency.

Project Workflow

CSV Dataset → SQL Server Data Import → Data Cleaning → Exploratory Data Analysis (EDA) → KPI Development → Power BI Dashboard Creation → Business Insights & Recommendations → GitHub Portfolio Documentation

Business Problem

Management requires visibility into revenue drivers, profitable product categories, customer purchasing behavior, regional performance, and shipping efficiency. This project was designed to provide data-driven insights to support strategic decision making.

SQL Analysis and Key Queries

Overall Business Performance: `SELECT SUM(Sales), SUM(Profit), SUM(Quantity) FROM superstore;`

Regional Performance: `SELECT Region, SUM(Sales), SUM(Profit) FROM superstore GROUP BY Region;`

Category Profitability: `SELECT Category, SUM(Profit) FROM superstore GROUP BY Category;`

Top Products by Revenue: `SELECT TOP 10 Product_Name, SUM(Sales) Revenue FROM superstore GROUP BY Product_Name ORDER BY Revenue DESC;`

Customer Segment Analysis: `SELECT Segment, SUM(Sales) Revenue FROM superstore GROUP BY Segment;`

Monthly Sales Trend: `SELECT YEAR(Order_Date), MONTH(Order_Date), SUM(Sales) FROM superstore GROUP BY YEAR(Order_Date), MONTH(Order_Date);`

Shipping Performance: `SELECT Ship_Mode, AVG(DATEDIFF(day, Order_Date, Ship_Date)) FROM superstore GROUP BY Ship_Mode;`

Key Findings

- Total revenue reached approximately \$2.3 million.

- The Consumer segment generated over half of total revenue and remains the most valuable customer segment.
- Technology products generated the highest profit while Furniture generated the lowest profit.
- The West region recorded the highest profitability among all regions.
- California emerged as the highest revenue-generating state.
- Standard Class shipping recorded the highest average shipping duration.
- Revenue and profit are concentrated among a relatively small number of products and markets.

Business Impact

Management can use these insights to allocate resources more effectively. Marketing efforts can focus on Consumer customers, inventory and promotional investments can prioritize Technology products, and operational teams can improve Standard Class shipping performance. Regional strategies used in the West can be replicated in lower-performing markets to drive profitability. California should remain a strategic focus area for customer retention and growth.

Future Improvements

- Develop customer lifetime value analysis.
- Introduce predictive sales forecasting models.
- Perform customer churn analysis.
- Build automated ETL pipelines using SSIS or Azure Data Factory.
- Implement role-based executive dashboards.
- Analyze seasonality and demand forecasting at product level.

Power BI Dashboard Screenshots

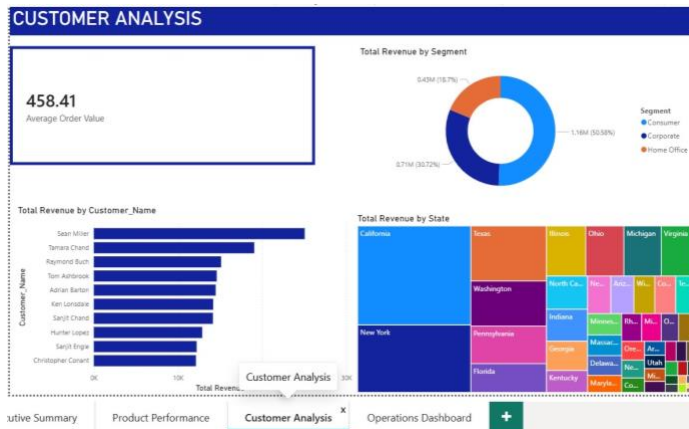
Executive Summary



Product Performance Dashboard



Customer Analysis Dashboard



Operations Dashboard

